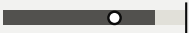


Why the policy runs at 100 Hz

The whole control loop on the robot's Raspberry Pi 3B, with the web interface running

Policy rate	Period	Compute budget used bar = web UI on, ○ = web UI off	Latency sensor to torque, mean	Jitter loop period, p99
50 Hz	20 ms	45 % 9.0 of 20 ms 	27.3 ms incl. 10.0 ms hold	+2.41 ms worst +29.3 ms
100 Hz deployed	10 ms	83 % 8.3 of 10 ms 	22.3 ms incl. 5.0 ms hold	+0.40 ms worst +2.3 ms
200 Hz	5 ms	166 % 8.3 of 5 ms 	19.8 ms* incl. 2.5 ms hold	+0.21 ms* worst +2.5 ms

Budget = policy tick inside the web-UI daemon (5.3–6.1 ms, midpoint; 79 %–87 % at 100 Hz) + safety governor 1.9 ms + CAN I/O 0.7 ms per 10 ms loop tick.

○ web UI stopped: the policy tick falls to 3.47 ms in a process of its own.

Latency = feedback wait 2.5 ms + drain to bus 8.3 ms + CAN transport 6.5 ms (measured, MIT sweep) + zero-order hold of half a period.

Jitter = loop-period overrun, Pi loop bench 2026-08-06, web UI running (light 0.3 ms policy).

* nominal: the 8.3 ms tick exceeds the 5 ms period (realised 120 Hz).